

Huzaifa Ahmad

+1 236-969-4649 | huzafaahmad@uvic.ca | Victoria, BC | Available January 2027 for a 4 to 8 month co-op term

EDUCATION

University of Victoria

Victoria, BC

Bachelor of Engineering in Computer Engineering

Expected Apr 2028

- **Relevant coursework:** Digital Design, Microprocessor Systems, VLSI Design, Signals & Systems, Electronic Circuit Analysis & Design, Analog Electronics

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, VHDL, MATLAB, Assembly

Frameworks & Tools: React, Git, Linux, FreeRTOS, MicroPython, Xilinx Vivado

Embedded & Digital: STM32 (Cortex-M), RP2040, Basys-3 FPGA, SPI, I2C, UART, ADC, PWM, bare-metal firmware

Hardware & Lab: KiCad, Altium Designer, schematic capture, PCB layout, surface-mount (SMD) soldering, board bring-up, oscilloscopes, multimeters

EXPERIENCE

UVic Satellite Design Club

Aug 2024 – Present

Hardware Engineer, On-Board Computer

Victoria, BC

- Contributed on-board computer hardware to MARMOTSat, a 3U ionospheric research CubeSat built by nearly 60 UVic contributors and launched on SpaceX's Transporter-17 in July 2026.
- Selected 40+ components by checking electrical, power-dissipation, thermal, and footprint requirements against manufacturer datasheets, finalizing the on-board computer's bill of materials (BOM) for production.
- Hand-assembled prototype boards by reflow soldering 0805, 0603, and SOIC parts, then verified power rails and signal continuity on the bench before handing them off for subsystem integration.
- Wrote a repeatable assembly and bring-up test procedure that other members used to test later board revisions.

UVic Rocketry

Jun 2023 – Present

Software Developer, Ground-Support Telemetry

Victoria, BC

- Contributed ground-support telemetry software to Andúril, UVic's Launch Canada rocket series, including Andúril-2, which placed 5th at Launch Canada 2025 after reaching Mach 2.2 and 32,694 ft.
- Developed the sensor-configuration UI for the team's ground-support telemetry app in React and TypeScript, used by the avionics team to set parameters for 6 onboard modules, including the inertial measurement unit (IMU), barometer, airbrakes, and strain gauges, during pre-launch testing.

PROJECTS

Custom PCB Design | Altium Designer

May – Jun 2026

- Produced a design-rule-check (DRC) clean manufacturing package with Gerbers, ODB++, BOM, and assembly drawings by taking a board through the full Altium Designer flow of schematic capture, placement, routing, and copper pours.

Pipelined CPU | VHDL, Xilinx Vivado, Basys-3 FPGA, Assembly

Jan – Apr 2026

- Designed a 16-bit, 5-stage pipelined CPU in VHDL with a custom instruction set, ALU, control unit, and hazard detection, validated with register-transfer level (RTL) testbenches.
- Met all static timing constraints at 50.8 MHz on a Basys-3 FPGA while using under 5% of the device's resources.

Real-Time OS Traffic Light Controller | C, FreeRTOS, STM32F4, SPI, ADC

Jan – Apr 2026

- Programmed a multi-task FreeRTOS application on an STM32F4 that drives 24 LEDs through 3 daisy-chained shift registers, using message queues and timers between tasks and ADC input to set cycle timing.
- Tuned task priorities and ran extended stress tests with no missed deadlines observed.

FM Radio-Clock PCB | KiCad, RP2040, MicroPython, SPI, I2C

Jan – Apr 2023

- Prototyped and brought up a hand-assembled 2-layer PCB with an RP2040, RDA5807 FM tuner, LM386 audio amplifier, SSD1306 OLED, and rotary-encoder controls on regulated 3.3 V and 5 V rails.
- Debugged the board with an oscilloscope and multimeter, then implemented MicroPython drivers over SPI and I2C for the clock display, alarms, FM tuning, and volume control.